

**SECTION 1. PRODUCT AND COMPANY IDENTIFICATION**

Product name : CHEM-STAR ACID PLUS

Other means of identification : Not applicable

Recommended use : Cleaning product

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.0 % - 0.1 %

Company : Ecolab Inc.  
1 Ecolab Place  
St. Paul, Minnesota USA 55102  
1-800-352-5326

Emergency health information : 1-800-328-0026 (US/Canada), 1-651-222-5352 (outside US)

Issuing date : 04/30/2019

**SECTION 2. HAZARDS IDENTIFICATION**
**GHS Classification**
**Product AS SOLD**

Skin corrosion : Category 1A  
Serious eye damage : Category 1

**Product AT USE DILUTION**

Skin corrosion : Category 1A  
Serious eye damage : Category 1

**GHS label elements**
**Product AS SOLD**

Hazard pictograms :



Signal Word : Danger

Hazard Statements : Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**

Wash skin thoroughly after handling. Wear protective gloves/ protective clothing/ eye protection/ face protection.

**Response:**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/doctor. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor. Wash contaminated clothing before reuse.

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

**Storage:**

Store locked up.

**Disposal:**

Dispose of contents/ container to an approved waste disposal plant.

**Product AT USE DILUTION**

Hazard pictograms

:



Signal Word

: Danger

Hazard Statements

: Causes severe skin burns and eye damage.

Precautionary Statements

: **Prevention:**

Wash skin thoroughly after handling. Wear protective gloves/ protective clothing/ eye protection/ face protection.

**Response:**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/doctor. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor. Wash contaminated clothing before reuse.

**Storage:**

Store locked up.

**Disposal:**

Dispose of contents/ container to an approved waste disposal plant.

**Product AS SOLD**

Other hazards

: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

### SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

**Product AS SOLD**

Pure substance/mixture

: Mixture

**Chemical name**

Phosphoric acid

sulfuric acid

nitric acid

**CAS-No.**

7664-38-2

7664-93-9

7697-37-2

**Concentration (%)**

10 - 30

1 - 5

1 - 5

**Product AT USE DILUTION**

No hazardous ingredients

### SECTION 4. FIRST AID MEASURES

**Product AS SOLD**

In case of eye contact

: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact

: Wash off immediately with plenty of water for at least 15 minutes. Use

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

- If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
- If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.
- Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.
- Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

#### Product AT USE DILUTION

- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
- If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
- If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

### SECTION 5. FIRE-FIGHTING MEASURES

#### Product AS SOLD

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : None known.
- Specific hazards during fire fighting : Not flammable or combustible.
- Hazardous combustion products : Decomposition products may include the following materials:  
Carbon oxides  
Nitrogen oxides (NOx)  
Sulfur oxides  
Oxides of phosphorus
- Special protective equipment for fire-fighters : Use personal protective equipment.
- Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

and/or explosion do not breathe fumes.

#### SECTION 6. ACCIDENTAL RELEASE MEASURES

##### Product AS SOLD

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

##### Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

#### SECTION 7. HANDLING AND STORAGE

##### Product AS SOLD

Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Conditions for safe storage : Keep away from strong bases. Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : -30 °C to 50 °C

##### Product AT USE DILUTION

Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

#### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

#### Product AS SOLD

##### Ingredients with workplace control parameters

| Components      | CAS-No.   | Form of exposure        | Permissible concentration | Basis     |
|-----------------|-----------|-------------------------|---------------------------|-----------|
| Phosphoric acid | 7664-38-2 | TWA                     | 1 mg/m3                   | ACGIH     |
|                 |           | STEL                    | 3 mg/m3                   | ACGIH     |
|                 |           | TWA                     | 1 mg/m3                   | NIOSH REL |
|                 |           | STEL                    | 3 mg/m3                   | NIOSH REL |
|                 |           | TWA                     | 1 mg/m3                   | OSHA Z1   |
| sulphuric acid  | 7664-93-9 | TWA (Thoracic fraction) | 0.2 mg/m3                 | ACGIH     |
|                 |           | TWA                     | 1 mg/m3                   | NIOSH REL |
|                 |           | TWA                     | 1 mg/m3                   | OSHA Z1   |
| Nitric acid     | 7697-37-2 | TWA                     | 2 ppm                     | ACGIH     |
|                 |           | STEL                    | 4 ppm                     | ACGIH     |
|                 |           | STEL                    | 4 ppm<br>10 mg/m3         | NIOSH REL |
|                 |           | TWA                     | 2 ppm<br>5 mg/m3          | NIOSH REL |
|                 |           | TWA                     | 2 ppm<br>5 mg/m3          | OSHA Z1   |

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : Wear eye protection/ face protection.

Hand protection : Wear the following personal protective equipment:  
Standard glove type.  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

#### Product AT USE DILUTION

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : Wear eye protection/ face protection.

Hand protection : Wear the following personal protective equipment:  
Standard glove type.

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

|                        |                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------|
|                        | Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.        |
| Skin protection        | : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing     |
| Respiratory protection | : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. |

### SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         | Product AS SOLD                                | Product AT USE DILUTION |
|-----------------------------------------|------------------------------------------------|-------------------------|
| Appearance                              | : liquid                                       | liquid                  |
| Color                                   | : clear, red                                   | colorless               |
| Odor                                    | : slight                                       | odorless                |
| pH                                      | : 0.5 - 0.9, (100 %)                           | 2.0 - 3.0               |
| Flash point                             | : Not applicable, Does not sustain combustion. |                         |
| Odor Threshold                          | : No data available                            |                         |
| Melting point/freezing point            | : No data available                            |                         |
| Initial boiling point and boiling range | : > 100 °C                                     |                         |
| Evaporation rate                        | : No data available                            |                         |
| Flammability (solid, gas)               | : No data available                            |                         |
| Upper explosion limit                   | : No data available                            |                         |
| Lower explosion limit                   | : No data available                            |                         |
| Vapor pressure                          | : No data available                            |                         |
| Relative vapor density                  | : No data available                            |                         |
| Relative density                        | : 1.105 - 1.135                                |                         |
| Water solubility                        | : soluble                                      |                         |
| Solubility in other solvents            | : No data available                            |                         |
| Partition coefficient: n-octanol/water  | : No data available                            |                         |
| Autoignition temperature                | : No data available                            |                         |
| Thermal decomposition                   | : No data available                            |                         |
| Viscosity, kinematic                    | : No data available                            |                         |
| Explosive properties                    | : No data available                            |                         |
| Oxidizing properties                    | : No data available                            |                         |
| Molecular weight                        | : No data available                            |                         |
| VOC                                     | : No data available                            |                         |

### SECTION 10. STABILITY AND REACTIVITY

#### Product AS SOLD

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| Reactivity         | : No dangerous reaction known under conditions of normal use. |
| Chemical stability | : Stable under normal conditions.                             |

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Possibility of hazardous reactions | : Do not mix with bleach or other chlorinated products – will cause chlorine gas.                                                                |
| Conditions to avoid                | : None known.                                                                                                                                    |
| Incompatible materials             | : Bases<br>Metals<br>Organic materials                                                                                                           |
| Hazardous decomposition products   | : Decomposition products may include the following materials:<br>Carbon oxides<br>Nitrogen oxides (NOx)<br>Sulfur oxides<br>Oxides of phosphorus |

### SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

#### Potential Health Effects

##### Product AS SOLD

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes serious eye damage.                                  |
| Skin             | : Causes severe skin burns.                                   |
| Ingestion        | : Causes digestive tract burns.                               |
| Inhalation       | : May cause nose, throat, and lung irritation.                |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

##### Product AT USE DILUTION

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes serious eye damage.                                  |
| Skin             | : Causes severe skin burns.                                   |
| Ingestion        | : Causes digestive tract burns.                               |
| Inhalation       | : May cause nose, throat, and lung irritation.                |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

#### Experience with human exposure

##### Product AS SOLD

|              |                                 |
|--------------|---------------------------------|
| Eye contact  | : Redness, Pain, Corrosion      |
| Skin contact | : Redness, Pain, Corrosion      |
| Ingestion    | : Corrosion, Abdominal pain     |
| Inhalation   | : Respiratory irritation, Cough |

##### Product AT USE DILUTION

|             |                            |
|-------------|----------------------------|
| Eye contact | : Redness, Pain, Corrosion |
|-------------|----------------------------|

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

|              |                                 |
|--------------|---------------------------------|
| Skin contact | : Redness, Pain, Corrosion      |
| Ingestion    | : Corrosion, Abdominal pain     |
| Inhalation   | : Respiratory irritation, Cough |

#### Toxicity

##### Product AS SOLD

##### Product

|                                   |                     |
|-----------------------------------|---------------------|
| Acute oral toxicity               | : No data available |
| Acute inhalation toxicity         | : No data available |
| Acute dermal toxicity             | : No data available |
| Skin corrosion/irritation         | : No data available |
| Serious eye damage/eye irritation | : No data available |
| Respiratory or skin sensitization | : No data available |
| Carcinogenicity                   | : No data available |
| Reproductive effects              | : No data available |
| Germ cell mutagenicity            | : No data available |
| Teratogenicity                    | : No data available |
| STOT-single exposure              | : No data available |
| STOT-repeated exposure            | : No data available |
| Aspiration toxicity               | : No data available |

##### Components

|                     |                                              |
|---------------------|----------------------------------------------|
| Acute oral toxicity | : Phosphoric acid<br>LD50 Rat: > 2,600 mg/kg |
|---------------------|----------------------------------------------|

##### Components

|                       |                                                 |
|-----------------------|-------------------------------------------------|
| Acute dermal toxicity | : Phosphoric acid<br>LD50 Rabbit: > 2,000 mg/kg |
|-----------------------|-------------------------------------------------|

### SECTION 12. ECOLOGICAL INFORMATION

##### Product AS SOLD

##### Ecotoxicity

|                       |                                                       |
|-----------------------|-------------------------------------------------------|
| Environmental Effects | : This product has no known ecotoxicological effects. |
|-----------------------|-------------------------------------------------------|

##### Product

|                                                     |                         |
|-----------------------------------------------------|-------------------------|
| Toxicity to fish                                    | : 96 h LC50: > 120 mg/l |
| Toxicity to daphnia and other aquatic invertebrates | : 48 h EC50: > 120 mg/l |
| Toxicity to algae                                   | : No data available     |

##### Components

|                   |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| Toxicity to algae | : Phosphoric acid<br>72 h EC50 <i>Desmodesmus subspicatus</i> (green algae): > 100 mg/l |
|-------------------|-----------------------------------------------------------------------------------------|



## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

#### Persistence and degradability

##### Product AS SOLD

Not applicable - inorganic

##### Product AT USE DILUTION

Not applicable - inorganic

#### Bioaccumulative potential

No data available

#### Mobility in soil

No data available

#### Other adverse effects

No data available

### SECTION 13. DISPOSAL CONSIDERATIONS

#### Product AS SOLD

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

RCRA - Resource Conservation and Recovery Authorization Act Hazardous waste : D002 (Corrosive)

#### Product AT USE DILUTION

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

### SECTION 14. TRANSPORT INFORMATION

#### Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

#### Land transport (DOT)

UN number : 3264  
Description of the goods : Corrosive liquid, acidic, inorganic, n.o.s.  
(Nitric acid, Phosphoric acid)  
Class : 8  
Packing group : III

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

Environmentally hazardous : no

#### Sea transport (IMDG/IMO)

UN number : 3264  
Description of the goods : CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.  
(Nitric acid, Phosphoric acid)  
Class : 8  
Packing group : III  
Marine pollutant : no

### SECTION 15. REGULATORY INFORMATION

#### Product AS SOLD

#### EPCRA - Emergency Planning and Community Right-to-Know

##### CERCLA Reportable Quantity

| Components    | CAS-No.   | Component RQ (lbs) | Calculated product RQ (lbs) |
|---------------|-----------|--------------------|-----------------------------|
| sulfuric acid | 7664-93-9 | 1000               | 2146                        |

##### SARA 304 Extremely Hazardous Substances Reportable Quantity

| Components    | CAS-No.   | Component RQ (lbs) | Calculated product RQ (lbs) |
|---------------|-----------|--------------------|-----------------------------|
| sulfuric acid | 7664-93-9 | 1000               | 21462                       |

**SARA 311/312 Hazards** : Acute Health Hazard

**SARA 302** : The following components are subject to reporting levels established by SARA Title III, Section 302:

|               |           |          |
|---------------|-----------|----------|
| sulfuric acid | 7664-93-9 | 4.6595 % |
| nitric acid   | 7697-37-2 | 4.489 %  |

**SARA 313** : The following components are subject to reporting levels established by SARA Title III, Section 313:

|             |           |         |
|-------------|-----------|---------|
| nitric acid | 7697-37-2 | 4.489 % |
|-------------|-----------|---------|

#### California Prop. 65

This product does not contain any chemicals known to the State of California to cause cancer, birth, or any other reproductive defects.

#### The ingredients of this product are reported in the following inventories:

##### United States TSCA Inventory :

On TSCA Inventory

##### Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL

##### Australia Inventory of Chemical Substances (AICS) :

On the inventory, or in compliance with the inventory

##### New Zealand. Inventory of Chemical Substances :

On the inventory, or in compliance with the inventory

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

**Japan. ENCS - Existing and New Chemical Substances Inventory :**

On the inventory, or in compliance with the inventory

**Japan. ISHL - Inventory of Chemical Substances (METI) :**

not determined

**Korea. Korean Existing Chemicals Inventory (KECI) :**

On the inventory, or in compliance with the inventory

**Philippines Inventory of Chemicals and Chemical Substances (PICCS) :**

On the inventory, or in compliance with the inventory

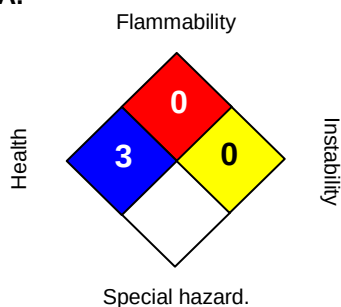
**China. Inventory of Existing Chemical Substances in China (IECSC) :**

On the inventory, or in compliance with the inventory

### SECTION 16. OTHER INFORMATION

#### Product AS SOLD

##### NFPA:



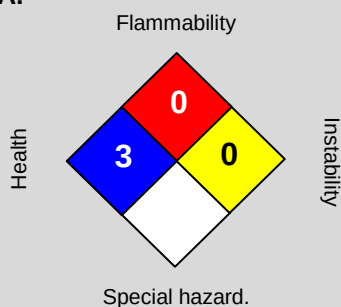
##### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 3 |
| FLAMMABILITY    | 0 |
| PHYSICAL HAZARD | 0 |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

#### Product AT USE DILUTION

##### NFPA:



##### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 3 |
| FLAMMABILITY    | 0 |
| PHYSICAL HAZARD | 0 |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

Issuing date : 04/30/2019  
Version : 1.3  
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is

## SAFETY DATA SHEET

### CHEM-STAR ACID PLUS

not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.